

MONK: A Cryptographic Governance Kernel for Trusted BVLOS Corridor Operations and UAS Traffic Management (UTM) Integration

Student Research Initiative

This research project is proposed by **Kevin Alexander Smith**, a student in the Cybersecurity Technology program at the **University of Maryland Global Campus (UMGC)** and a **Federal Aviation Administration (FAA) Part 107 Remote Pilot, Project Management Professional (PMP), and Certified Information Systems Security Professional (CISSP)** with more than three decades of experience leading cybersecurity and infrastructure programs supporting critical enterprise and government systems.

Project at a Glance

Student Investigator:

Kevin Alexander Smith
Cybersecurity Technology Program
University of Maryland Global Campus (UMGC)

Certifications:

FAA Part 107 Remote Pilot | Project Management Professional (PMP) | Certified Information Systems Security Professional (CISSP)

Research Focus:

Cryptographic telemetry governance for Beyond Visual Line of Sight (BVLOS) drone corridor operations.

Prototype Deliverable:

Flight-testable **Minimum Viable Product (MVP)** MONK governance payload.

Internship Environment:

University of Maryland UAS Research and Operations Center (UROC)
Chesapeake UAS Route Network (CURN)

Mentors:

Dr. Alexander Kott, PhD – Artificial Intelligence and Cybersecurity Subject Matter Expert
Michael (Mike) Ducker – Director, Maryland Energy Innovation Accelerator (MEIA)
Emily Sheppard – Program Leadership, Maryland Energy Innovation Accelerator (MEIA)
Lauren Segura – University of Maryland Global Campus (UMGC)

Abstract

This project investigates a hardware-anchored governance framework for autonomous drone corridor operations within the **Chesapeake UAS Route Network (CURN)** operated by the **University of Maryland UAS Research and Operations Center (UROC)**. The research centers on the **Modular Operational Network Kernel (MONK)**, a modular architecture designed to verify compliance of **Beyond Visual Line of Sight (BVLOS)** missions through deterministic validation of flight telemetry and secure cryptographic logging.

The internship will produce a **flight-testable Minimum Viable Product (MVP)** prototype of the MONK governance module capable of validating telemetry and generating tamper-resistant compliance records during UAS operations. The prototype will demonstrate a constrained but operational implementation suitable for experimental evaluation within the **10-week UROC Summer Internship Program**.

The system will interface with a drone flight controller using **MAVLink telemetry** and process flight data through an edge computing module. A Raspberry Pi-based processor will evaluate telemetry against pre-authorized mission geofences and corridor constraints. Validated telemetry will then be cryptographically signed using a **Trusted Platform Module (TPM)**, producing tamper-resistant flight logs capable of supporting compliance auditing and corridor governance.

Validated telemetry records will also generate a cryptographically verifiable compliance token referred to as the **MONK-Key**. The MONK-Key represents a signed proof of operational compliance that could be consumed by future **UAS Traffic Management (UTM)** systems, enabling trusted verification of corridor operations by airspace governance platforms.

The prototype will operate as an **observational governance payload**, monitoring telemetry and recording compliance data without modifying aircraft flight control systems. The project will leverage the investigator's **FAA Part 107 operational experience and PMP project management methodology** to deliver a controlled MVP prototype within the internship schedule.

The research also explores how cryptographically verified telemetry records produced by MONK could support future integration with emerging **UAS Traffic Management (UTM)** systems by providing trusted flight compliance data for corridor operators and airspace governance platforms.

Research Objective

Evaluate whether a modular cryptographic telemetry governance layer can improve **trust, accountability, and compliance verification** for BVLOS drone operations within structured airspace corridors.

Experimental Evaluation Metrics

The MONK MVP prototype will be evaluated using measurable operational metrics appropriate for a 10-week research cycle.

Telemetry Integrity Verification

Percentage of MAVLink telemetry packets successfully validated against mission constraints.

System Latency

Processing time between telemetry ingestion and compliance validation output.

Tamper Resistance

Ability of the TPM-anchored log chain to detect or prevent unauthorized modification of recorded flight data.

Mission Geofence Compliance

Accuracy of deterministic validation of drone position relative to authorized corridor boundaries.

Flight Anomaly Detection

Identification and logging of telemetry conditions associated with safety-critical events (e.g., corridor violations, GNSS inconsistencies, or loss-of-link scenarios) that would normally trigger return-to-home or emergency landing procedures.

Technical Architecture (Conceptual)

Drone Flight Controller

↓ (MAVLink Telemetry)

MONK Governance Payload

(Raspberry Pi Edge Processor)

Telemetry Validation Engine

↓

Mission Constraint / Geofence Engine

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Trusted Platform Module (TPM) Cryptographic Signing

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MONK-Key (Signed Compliance Token)

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UAS Traffic Management (UTM)

or Corridor Governance Systems

Technical Implementation

The prototype governance payload will consist of:

- MAVLink telemetry ingestion via UART interface to the flight controller
- Edge processing on a Raspberry Pi-based module
- Python-based governance kernel performing deterministic telemetry validation
- Trusted Platform Module (TPM) for cryptographic signing and tamper-resistant logging

The module will operate alongside the flight controller as a **non-intrusive governance layer** designed to observe and verify operational compliance.

Student Investigator

Kevin Alexander Smith

Student — Cybersecurity Technology Program
University of Maryland Global Campus (UMGC)

Relevant credentials include:

Federal Aviation Administration (FAA) Part 107 Remote Pilot Certificate
Project Management Professional (PMP)
Certified Information Systems Security Professional (CISSP)

Kevin brings more than **33 years of experience in cybersecurity, infrastructure, and cloud transformation leadership**, including senior cybersecurity and infrastructure program management roles supporting global enterprise, healthcare, and government systems .

Current UMGC coursework includes:

Emerging Technologies in Drones and Autonomous Systems (DRON 315)
Generative Artificial Intelligence (ARIN 340) .

Mentorship and Advisory Support

Dr. Alexander Kott, PhD

Artificial Intelligence and Cybersecurity Subject Matter Expert

SCORE Mentor

Email: alexander.kott@scorevolunteer.org

LinkedIn: <https://www.linkedin.com/in/alexander-kott-a15515144>

Profile: <https://www.score.org/washingtondc/profile/alexander-kott>

Role: Academic advisor and technical subject matter expert supporting artificial intelligence, cybersecurity, and autonomous system governance.

Lauren Segura

University of Maryland Global Campus (UMGC)

Role: Academic coordination and program alignment supporting internship integration with UMGH coursework and student research objectives.

Michael (Mike) Ducker

Director

Maryland Energy Innovation Accelerator (MEIA)

Role: Industry mentor providing guidance on advanced air mobility ecosystems and commercialization strategy.

Emily Sheppard

Program Leadership

Maryland Energy Innovation Accelerator (MEIA)

Role: Innovation program advisor supporting technology acceleration and ecosystem integration.

Deliverables

At the conclusion of the internship the project will produce:

- Version 1.0 **flight-testable Minimum Viable Product (MVP)** MONK governance module
 - Telemetry validation and performance testing results
 - Flight anomaly detection observations
 - Final technical research report evaluating cryptographic governance layers for BVLOS corridor operations
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Policy and Research Context

This research contributes to emerging frameworks supporting safe integration of unmanned aircraft systems into the **National Airspace System**, complementing broader initiatives emphasizing trusted autonomy, telemetry integrity, and governance transparency for **Beyond Visual Line of Sight (BVLOS)** operations.

By exploring a cryptographic governance layer for BVLOS corridor operations, this research supports Maryland's broader leadership in **Advanced Air Mobility (AAM)** and contributes to the development of trusted operational frameworks necessary for safe, scalable integration of autonomous aircraft into the National Airspace System.